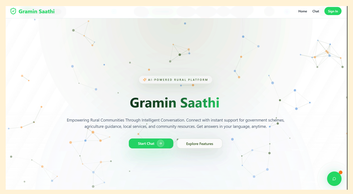
5TH SEM EL

Tech Stack- Web + PWA App



clerk · TypeScript · Tailwind CSS · React · shadcn/ui · Vite

Data

Pinecone — VectorDB

redis — Cache

Tech Stack- Backend

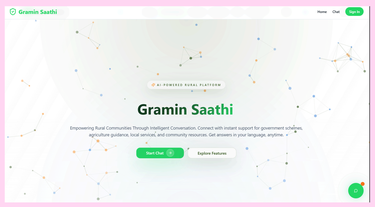
FastAPI · Gemini · LlamaIndex · Gemini Embedding · DuckDuckGo

Deployment

Vercel · Vertex AI · Render

Technology Stack

Web Interface



Web Dashboard

- Adaptive Chat UI – Renders real-time "thinking steps" and dual-language responses.
- Voice Overlay – Full-screen modal with particle effects for hands-free interaction.
- Smart Sessions – Auto-titles chats and isolates history via Clerk.

Tech Stack- Frontend

TypeScript · Tailwind CSS · React

Deployment

Vercel · Vertex AI · Render

Storage

redis — Cache

- Query Caching - Fast retrieval of frequent searches
- Local Memory / SQLite (via langcache)
- Monitoring Service - Latency tracking and hit-rate analysis

Pinecone — Knowledge Graph

- Vector Embeddings – Encodes scheme guidelines and scraped content.
- Semantic Search – Matches user queries to relevant documents via meaning.
- Relevance Scoring – Filters results to decide between internal data or web search.

API Layer

API Gateway · Document Processing · RAG System · Agent Planner... · User Auth... · Response Generation...

- RESTful Endpoints – Manages the primary chat interface via POST stream.
- Intelligent Routing – Orchestrates traffic between Cache, RAG, and Web Search (DuckDuckGo).
- Buffered Streaming – Accumulates and streams LLM responses.
- Identity Management – Integrates with Clerk to enforce authentication .

Backend Stack

FastAPI · LlamaIndex · Gemini

Data Source...

- Government Websites (HTML), Scheme PDFs (Yojanas) Scraper Service (scraper_service.py) Ingestion

Document Ingestion

Synchronization
- Scheduled document collection
- Batch processing workflows

Document Processsing...
- Document format conversion
- Text extraction and cleaning

Chunking and Embedding...
- Document segmentation
- Metadata extraction

Knowledge Graph Constructio...
- Entity extraction and linking
- Relationship mapping between documents

FastAPI · DuckDuckGo · sarvam.ai · @gov.in · PDF · Gemini Embedding

Hybrid Retriveval (RAG)

Hybrid Search
- Vector + keyword + graph search

Context Builder
- Top-K chunk selection
- Context window optimi...

Reranking
- Result refinement
- Relevance scoring

LlamaIndex